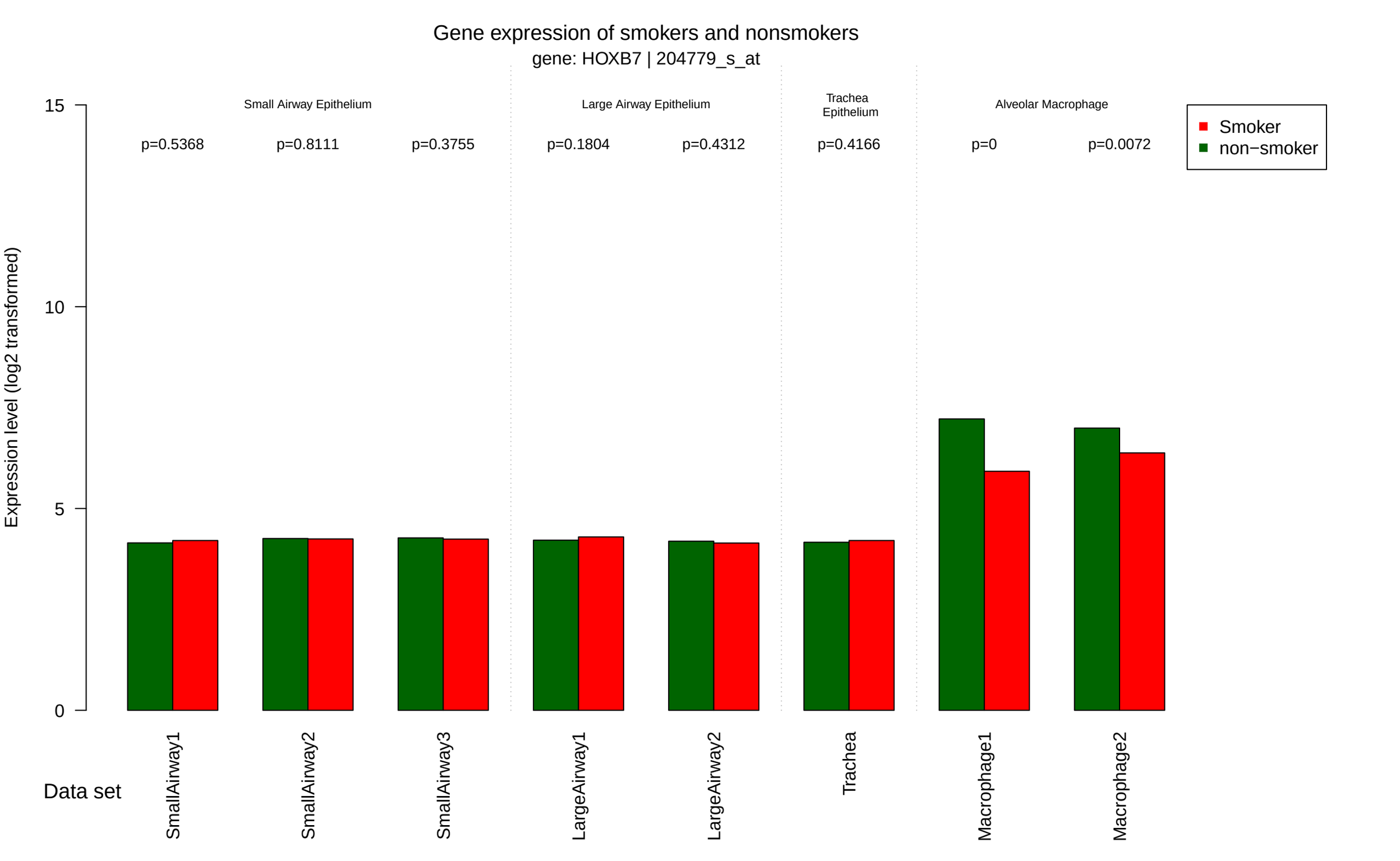
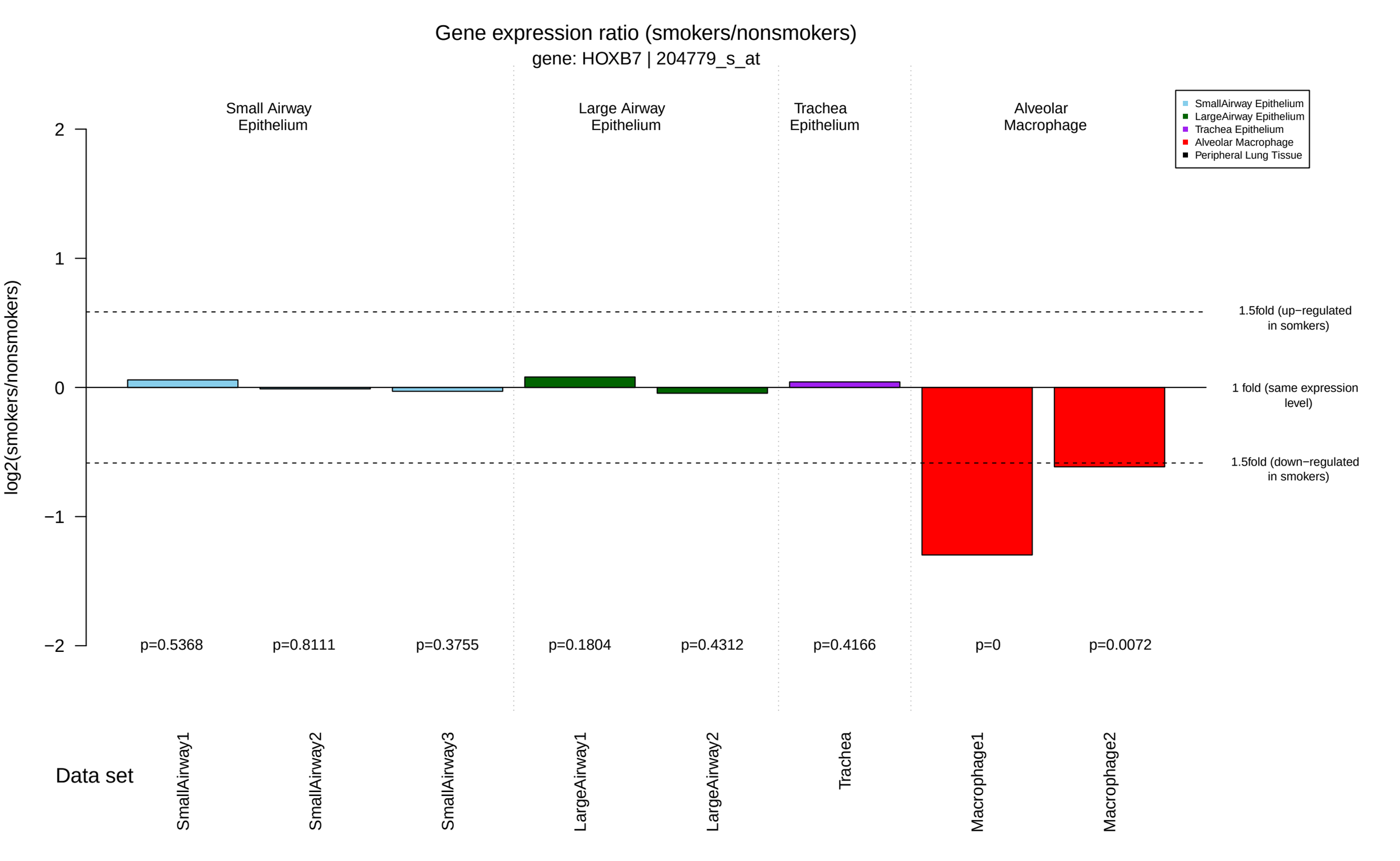
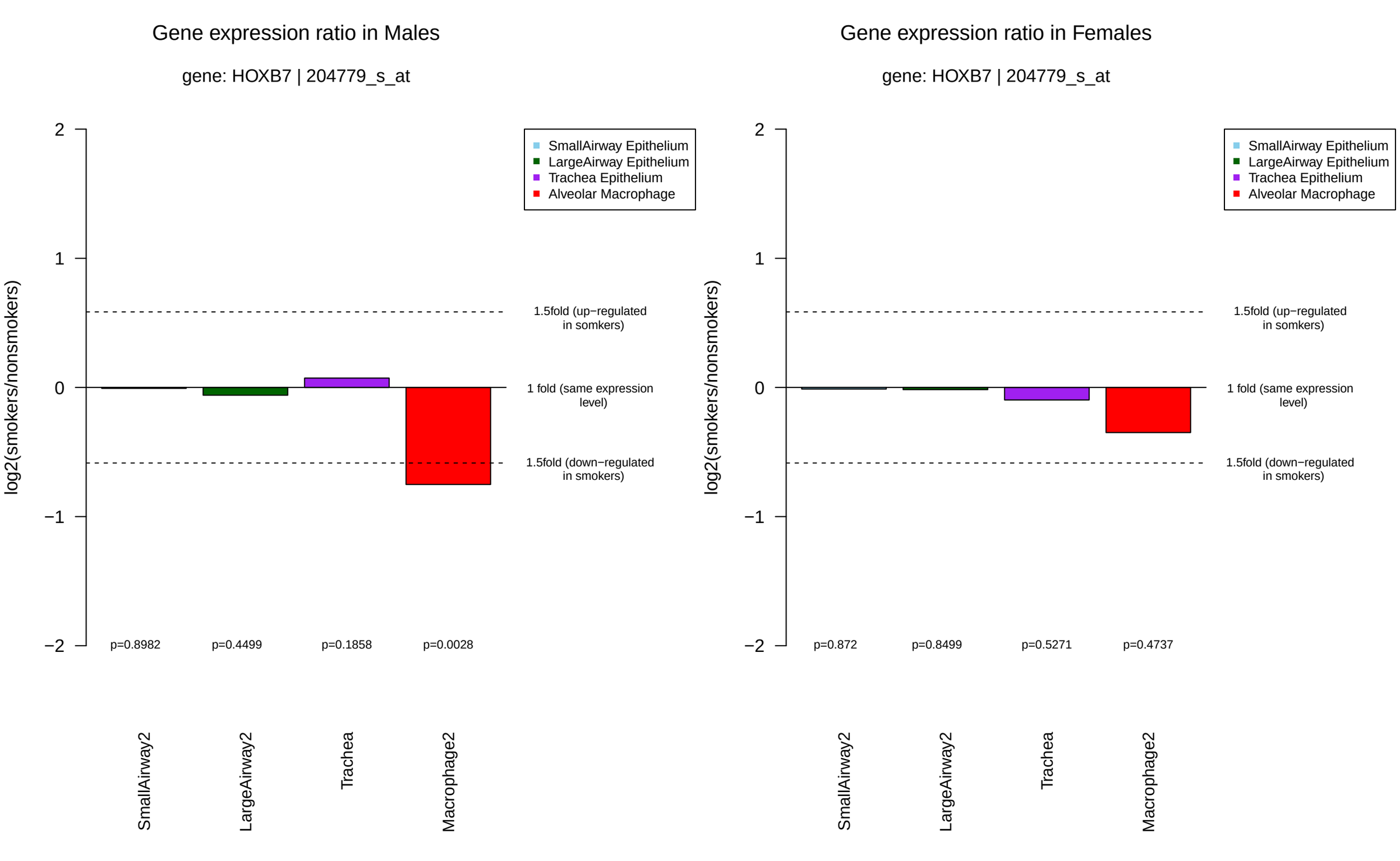
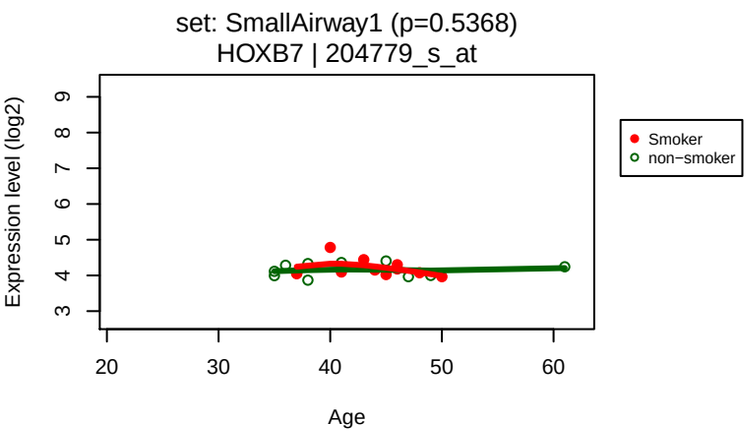
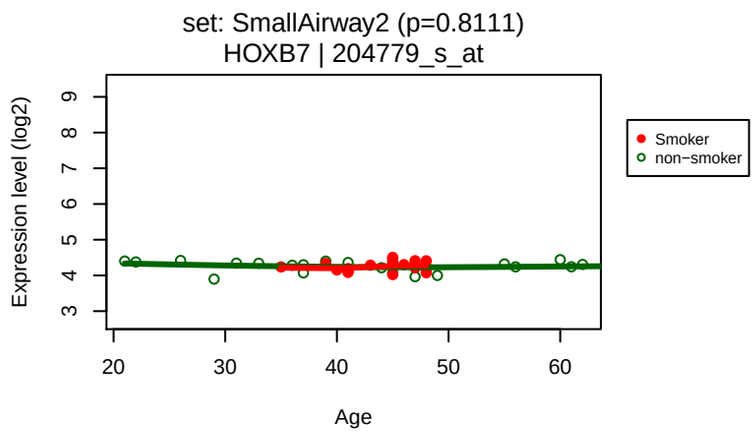
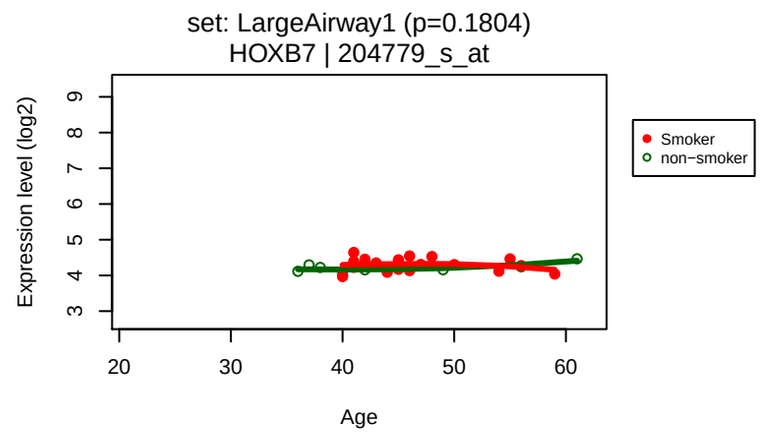

Gene expression of smokers and nonsmokers
gene: HOXB7 | 204779_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5368
p=0.8111
p=0.3755
p=0.1804
p=0.4312
p=0.4166
p=0
p=0.0072
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

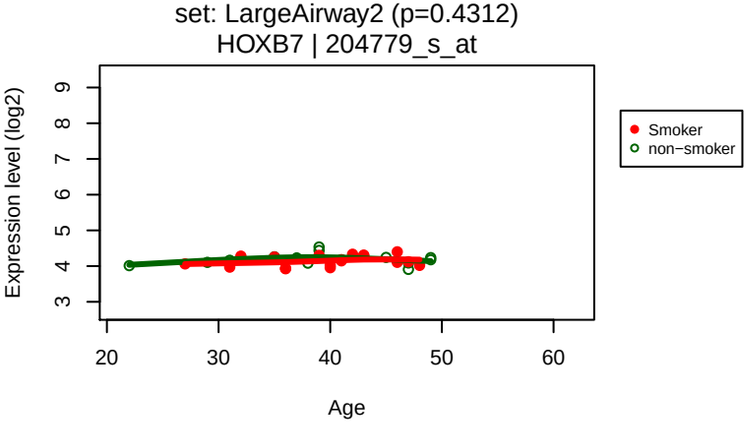

Gene expression ratio (smokers/nonsmokers)
gene: HOXB7 | 204779_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5368
p=0.8111
p=0.3755
p=0.1804
p=0.4312
p=0.4166
p=0
p=0.0072
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

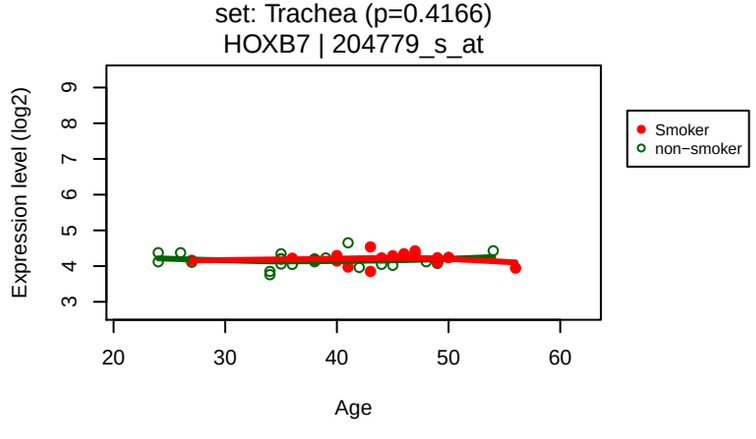

Gene expression ratio in Males
gene: HOXB7 | 204779_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8982
p=0.4499
p=0.1858
p=0.0028
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HOXB7 | 204779_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.872
p=0.8499
p=0.5271
p=0.4737
SmallAirway2
LargeAirway2
Trachea
Macrophage2

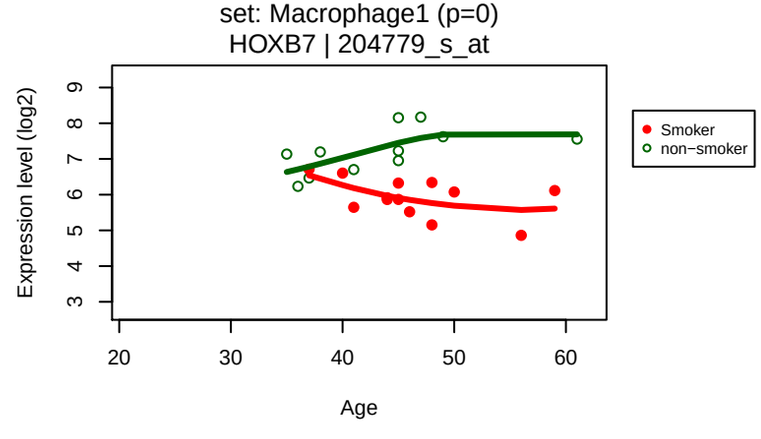

set: SmallAirway1 (p=0.5368)
HOXB7 | 204779_s_at
Expression level (log2)
Age
Smoker
non−smoker

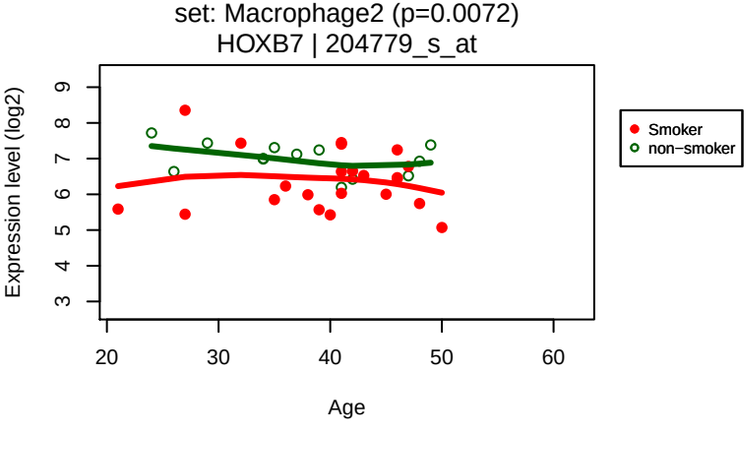

set: SmallAirway2 (p=0.8111)
HOXB7 | 204779_s_at
Expression level (log2)
Age
Smoker
non−smoker

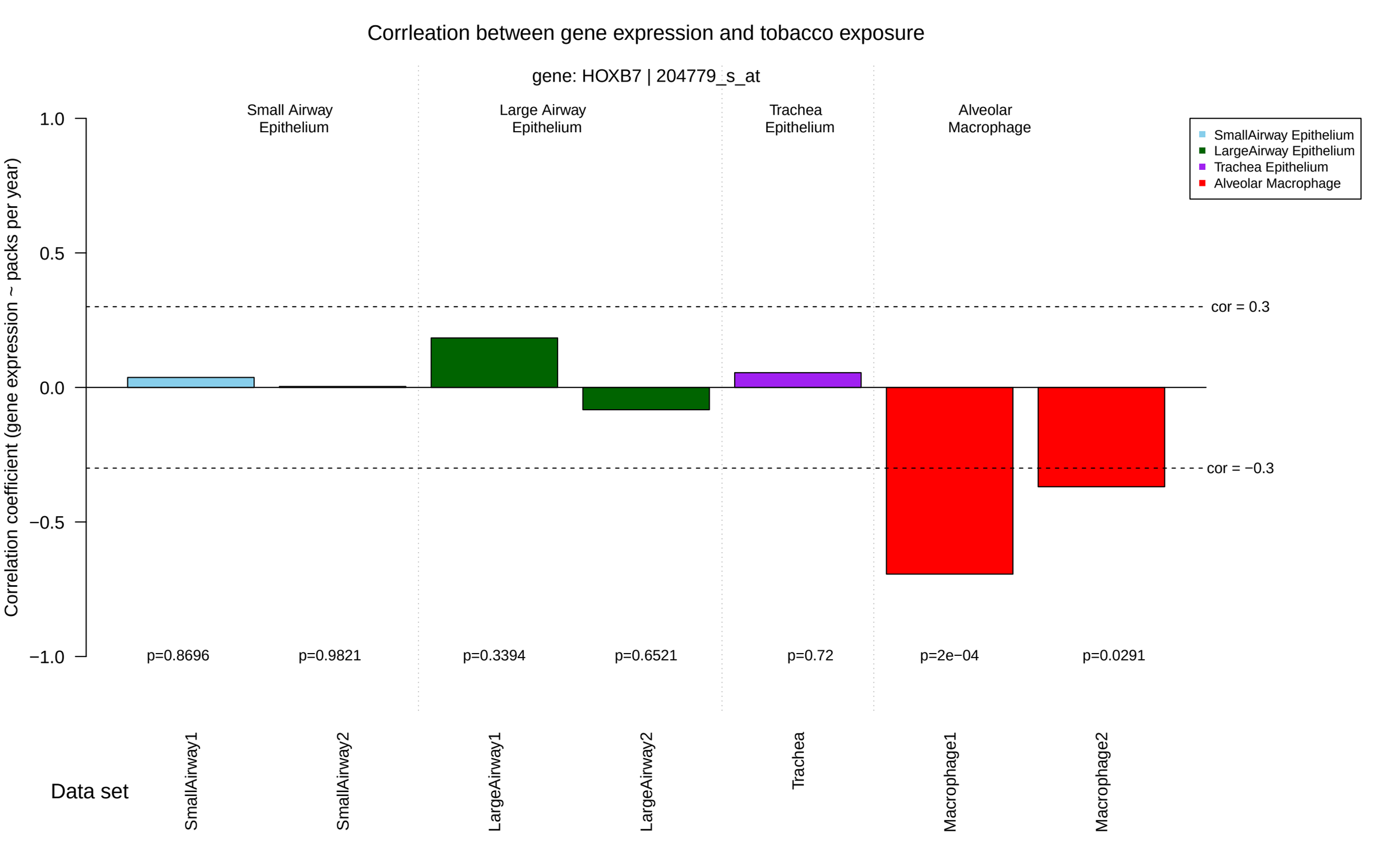

set: LargeAirway1 (p=0.1804)
HOXB7 | 204779_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4312)
HOXB7 | 204779_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4166)
HOXB7 | 204779_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0)
HOXB7 | 204779_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0072)
HOXB7 | 204779_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HOXB7 | 204779_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8696
p=0.9821
p=0.3394
p=0.6521
p=0.72
p=2e−04
p=0.0291
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2